

Wave W006 — Two-Path Decision

Project: MINERVA QCELL/RFCELL P&ID · **Status of prior waves:** W001–W005 complete (PR #546 merged, PR #547 open)
Pick one path below; W006 executes the selected option end-to-end.

Context (why this decision exists)

W005 reconciled the **design nomenclature register** (Excel, 97 tags) against the **as-drawn catalog** (SVG-instance, 141 real tags) and found **0 % exact tag overlap** — the two registers use orthogonal numbering schemes:

	Design register (Excel)	As-drawn catalog (SVG)
Numbering	Circuit-sequential (CV001 , TT100)	SVG-instance (CV560 , TT514)
Has location?	Free-text only ("40K circuit to pumping line")	X/Y coordinates + colour-class line (A/B/D/S/V/W) + layer
Has circuit/temp?	Embedded in location text (40 K / 4.5 K / 2 K)	Encoded via colour class + legend

The W005 report’s **#1 recommendation** was to build a **Design ↔ As-Drawn Cross-Map**.

Option A — Design ↔ As-Drawn Cross-Map ★ RECOMMENDED

What it solves

Turns the 0 % exact-overlap finding into a **functional bidirectional mapping** (CV001 ↔ CV560), enabling traceability from design intent → physical implementation. Foundation for commissioning, maintenance, and any future digital-twin work.

Approach (grounded in the actual W005 data)

- Feature extraction** per tag: ISA TYPE, circuit/temperature (parsed from Excel location text vs. catalog colour-class→legend), line assignment, and — catalog side only — X/Y + layer.
- Type-partitioned candidate matching** with **confidence scoring**:
 - TYPE match (mandatory gate)
 - Circuit/temperature-band agreement (40 K / 4.5 K / 2 K ↔ colour class)
 - Sequence-order heuristic within a type+circuit group (design N-th ↔ as-drawn N-th)
 - Manual validation seed (e.g. the PPT-confirmed TT535 / TT525 → PZ allocations)

3. **Confidence tiers:** HIGH (type+circuit+order), MEDIUM (type+circuit), LOW (type only — flagged for review).

Honest limitation (will be stated in the report, not hidden)

The design register has **no coordinates**, so a true spatial position match is impossible. Matching relies on TYPE + circuit-band + within-group ordering. Expect a meaningful share of MEDIUM/LOW pairs

that need engineering sign-off — we will **not** fabricate HIGH-confidence matches.

Deliverables

- `data/crossmap/design_to_asdrawn.json` (bidirectional mapping + reverse index)
- `data/crossmap/crossmap_confidence.json` (per-pair score + reasons)
- `reports/W006_CROSSMAP_REPORT.md` (validation report: tiered tables + unmapped list + recommendations)
- `reports/W006_crossmap_statistics.json` (coverage summary)
- `data/excel/canonical_register_v2.yaml` (canonical register + cross-references)
- `src/abacus_svg_pid/build_w006_crossmap.py` (reproducible engine, wired into `make.sh`)
- `tests/test_w006_crossmap.py` (bidirectional lookup, confidence bounds, golden seeds)
- Governance updates (`wave_status.json` , `CAPABILITY_MATRIX.md` , `wave_registry.json`)

Effort / impact

1 wave · **HIGH** engineering value · **blocks** future waves (W007-W009 traceability builds on this).

Option B — Interactive UI Viewer (originally planned W006)

What it provides

A standalone, browser-based P&ID navigator over the existing layered model.

Approach

1. **Layer toggle controls** — the 21-layer hierarchy with checkboxes (heat loads, spec dots, signals...).
2. **Pan / zoom / search** — SVG viewport manipulation; search by tag (`CV001` → highlight + zoom).
3. **Metadata popups** — click an element → show catalog/reconciliation details.
4. **Comparison views** — side-by-side QCELL vs RFCELL; colour-class isolation.

Deliverables

- `publish/interactive_viewer.html` (standalone, no server required)
- JavaScript layer-toggle + search/filter engine
- Metadata popups (instrument details from the catalog/canonical register)
- Per-layer / per-view PNG-PDF export controls
- `tests/test_w006_viewer.py` (data-binding + asset-integrity checks)

Effort / impact

1 wave · **MEDIUM** value · **blocks nothing** (nice-to-have; can be deferred to W007/W008).

Comparison

Criterion	A: Cross-Map	B: Interactive UI
Engineering priority	★★★★ HIGH (W005 #1 rec)	★★★ MEDIUM
Depends on	W005 canonical register	W003/W004 layer hierarchy
Complexity	Medium (heuristics + validation)	Medium (UI/UX + SVG manipulation)
Deliverable type	Data model + report	Interactive application
Stakeholders	Engineers, commissioning	All stakeholders, demos
Blocks future work?	Yes (W007–W009)	No
Can defer?	No (critical gap)	Yes

Recommendation

Execute Option A (Cross-Map) first — it closes the critical traceability gap identified in W005 and is a prerequisite for the later waves. Option B can follow in a subsequent wave.

User choice

- ☐ **Option A** — Design ↔ As-Drawn Cross-Map (recommended)
- ☐ **Option B** — Interactive UI Viewer
- ☐ **Both** (sequential; longer)
- ☐ **Other** (tell me)